

Luis Andrés Alemán Sánchez

Robotics Systems Engineer | Embedded Firmware | Industrial Automation

León, GTO, México • andresaleman142@gmail.com • andresalemn.github.io
linkedin.com/in/andresalemn

Summary

Robotics systems engineer bridging embedded firmware, control theory, and industrial automation — comfortable working end-to-end, from mechanical/electrical bring-up to real-time C++ control loops. Two years building torque and position controllers, CAN/CANopen distributed actuator networks, and ROS 2 robotic systems, alongside PLC and pneumatic automation for industrial clients. Native Spanish, professional English (C1).

Experience

Robotics & Control Engineer Jr.

León, Guanajuato, Mexico

Inbiodroid

May 2024 – present

- Developed torque-based joint controllers in C++ for a 20-DOF humanoid research platform, achieving sub-millimeter positioning accuracy via encoder feedback and custom control algorithms running in real time.
- Built and deployed embedded firmware on Teensy 4.1 and ESP32 microcontrollers across a distributed CAN/CANopen actuator network, covering real-time motion control, safety monitoring, and automated homing routines.
- Translated dynamics and kinematics research into custom torque and inverse-kinematics controllers for real-time teleoperation, integrating IMU-based motion capture over an existing UDP communication layer.
- Led system integration, calibration, and debugging across robotic hardware and software, plus PLC control logic (Siemens LOGO!) and electrical/pneumatic design for industrial automation projects.

Projects

3-DOF DC Motor Manipulator

University Project

- Designed and implemented embedded motor-control firmware in C++ for a 3-DOF robotic manipulator.
- Integrated quadrature encoders and tuned PID controllers to achieve ± 1 mm positioning accuracy.
- Developed a serial communication interface between the embedded controller and a Python application for monitoring and control.

Alexa-Controlled Robot (ROS 2 + Flask)

Personal Project

- Developed a ROS 2-based robotic control system integrating Amazon Alexa voice commands through a Flask web service and secure tunneling.
- Implemented ROS 2 nodes, lifecycle management, and ros2_control interfaces for real-time servo actuation and state feedback.
- Designed APIs and communication pipelines between cloud services and physical robotic hardware.

Blind Spot Detection System

Bachelor's Capstone Project

- Developed a Raspberry Pi driver-assistance prototype using Python, OpenCV, and ultrasonic sensing.
- Implemented signal-processing and digital-filtering techniques to improve obstacle-detection robustness and reduce sensor noise.
- Integrated computer-vision and distance-sensing modules into a single embedded application.

Skills

Programming: C, C++, Python, MATLAB

Robotics Software: ROS 2, ros2_control, RViz, TF2, URDF, MoveIt, Gazebo, Coppeliasim

Embedded Systems: Teensy, ESP32, Arduino, Raspberry Pi, Firmware Development, Real-Time Systems, CANopen, UART, SPI, I²C, UDP, Serial Communication

Actuation & Motor Control: ZeroErr eRob, CubeMars AK Series, MyActuator RMD, CAN/CANopen

Control & Modeling: PID Control, Kinematics, Dynamics, Inverse Kinematics, System Modeling, MATLAB/Simulink

Industrial Robotics: ABB, FANUC, Universal Robots, Mitsubishi

Industrial Automation: Siemens LOGO!, CODESYS, SolidWorks Electrical, FluidSIM, Electrical Schematics, Pneumatic Systems

CAD & Mechanical Design: SolidWorks

Tools: Git, GitHub, Docker, Linux (Ubuntu), VS Code, PlatformIO, LaTeX

Languages: Spanish (Native), English (C1), Finnish (A2)

Education

Universidad Anáhuac Mayab

BS in Mechatronics Engineering

Mérida, Yucatán, Mexico

Sept 2019 – Dec 2023

Universidad Anáhuac Mayab

Diploma in Cultural Management

Mérida, Yucatán, Mexico

Jan 2020 – May 2023

Completed undergraduate coursework in the Culmen Leadership and Excellence in Arts & Culture Program.

Certificates

Industrial Robotics Manufacturing Systems Integrator

Sept 2026

- State-issued competency certification (Secretaría de Economía de Guanajuato) in industrial robot integration across ABB, FANUC, Universal Robots, and Mitsubishi platforms.
- Covers architecture selection, safety risk analysis, and production-cell commissioning.

Complementary Education

- **Professional Development:** Specialized courses in ROS 2, MoveIt, Modern Robotics, Collaborative Robotics in Industry, Operating Systems, and Computer Networks (Coursera, Udemy, edX). Motion Planning — Invited participant in a graduate-level course at CIMAT (2026).
- **International Experience:** Selected to participate in academic exchange programs in Finland (2017–2018) and Canada (2022).